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Top Skills

Flask
DevOps
Educational Programs

Languages

Spanish (Limited Working)
Latin (Elementary)
English (Native or Bilingual)
Japanese (Elementary)
Chinese (Native or Bilingual)

Honors-Awards

USACO Silver (Java)
Microsoft Hunt the Wumpus Seattle First Place
AIME Qualification
Public Forum Debate District Champion, State Third Place
Distinction Award

Ryan Li

Undergraduate Student at CMU pursuing a B.S. in Computer Science

Kenmore, Washington, United States

Summary

Current undergraduate student pursuing a Bachelor's of Science in Computer Science with a strong, diverse passion in math, programming, computer-aided design, and debate. Proficient in many programming languages and varieties of handicraft. Loves to help others and lead them through the process of discovery. Interested in machine learning, logic, software development, and security and privacy. Always seeking the thrill of everything coming together in research.

Programming Languages: Java, Kotlin, Nix, Python, Rust, C#, TypeScript, HTML/CSS, SQL(PostgreSQL), LaTeX, C, RFrameworks: Django, Flask, Bootstrap, Material-UI, Node.js, ReactDeveloper Tools: Git, Google Cloud Platform, Visual Studio, VSCode, PyCharm, IntelliJ, Microsoft AzureLibraries: Tensorflow, SkLearn, HuggingFace, pandas, NumPy, Matplotlib, Pillow, Mozilla-Django OIDCCAD Tools: Fusion 360, Inventor, Blender, Adobe IllustratorPhysical Tools: 3D Printers, Resin Printers, Laser Cutters, Metal CNC Machines, Power Tools

Experience

Carnegie Mellon University

9 months

Research Assistant, Fake Docs project

July 2026 - Present (1 month)

Pittsburgh, PA

- Conducting labeling and machine learning work on fraudulent website detection

- Optimizing labeling speed for self and others using HCI intuition to designate hotkeys and UI/UX behavior

Research Assistant, SMS Phishing project

November 2025 - Present (9 months)

Pittsburgh, PA

- Conducting full-stack design for an application that conducts AI-generated warnings to detect scam messages, assisting Dr. Lorrie Cranor and Dr. Enze Liu by implementing experimental software for a large-scale study.
- Leading all technical development for the platform, translating research hypotheses into scalable Android features and managing the integration of LLM responses into the mobile environment.
- Implementing a secure module within a open-source SMS application to facilitate real-time scam detection.
- Engineered a robust, encrypted transmission system for user data to be sent to a LLM-powered backend via RESTful API.

ScottyLabs

11 months

DevOps and Internal Tooling Tech lead

April 2026 - Present (4 months)

Pittsburgh, PA

- Using NixOS to host and grant various data modules to diverse projects throughout all of ScottyLabs
- Building various internal applications to improve internal communications and organizational observability
- Acted immediately to fix VM crashes that caused outages for more than 3,000 users.
- Used Rust and declarative programming to develop applications that manage and standardize the backend of new projects
- Conducted a large-scale migration of the majority of past systems to more secure and optimized alternatives.
- Leading a small team, distributing tasks among members to achieve maximal efficiency and application of skills, while interacting with all sectors of ScottyLabs to understand their hosting needs.

Devops Tech Lead

November 2025 - April 2026 (6 months)

- Using NixOS to host and grant various data modules to diverse projects throughout all of ScottyLabs
- Conducting a large-scale migration of the majority of past systems to more secure and optimized alternatives.
- Leading a small team, distributing tasks among members to achieve maximal efficiency and application of skills, while interacting with all sectors of ScottyLabs to understand their hosting needs.

Devops Member

September 2025 - November 2025 (3 months)

Using Rust to develop applications that manage and standardize the backend of ScottyLabs projects

Leading a small team, distributing tasks among members to achieve maximal efficiency and application of skills.

MaxMyPoint

1 year 8 months

AI Engineer

December 2024 - Present (1 year 8 months)

Austin, TX

- Developing a LangGraph + PydanticAI + AWS AgentCore chatbot to assist over 56,000 users in effectively navigating the website and suggesting optimal travel plans.
- Engineered an MCP server to allow the chatbot to conduct operations for users, considering their account level and relevance of question
 - Conducting repeated prompt engineering and fine-tuning on DeepSeek and OpenAI models to ensure safety, consistency, and accuracy.
 - Storing information via AWS AgentCore Memory to track user questions, frequency, and history, ensuring better UX
 - Coordinating and working within a small team on front-end and back-end choices.
 - Communicating with leaders to establish wishes, features, and next steps for project.

Data Scientist

December 2025 - July 2026 (8 months)

Utilizing PostgreSQL databases to track hotel alert success rates based on type, hotel, and time

Creating visually effective data pages to analyze the effectiveness of hotel point redemptions, as well as an API to utilize it.

Engineering automatic workflows to conduct regular data analysis for continued accuracy

MaxFHR

Full Stack Engineer

March 2026 - May 2026 (3 months)

Schengen Area

- Created the full stack of a map display system for hotels in MaxFHR, designing caches to increase loading speed over multiple operations and gearing its frontend experience to optimize navigation and intuitiveness.

Eastside Preparatory School

Mentor, EPS FTC Robotics Teams

September 2025 - February 2026 (6 months)

- Remotely mentoring 3 teams totaling 30 people: FTC Teams 8103 (null robotics), 9884 (void robotics), and 15203 (undefined robotics)
- Advising students grades 8-12 on design choices and project/team management skills.

Eastside Preparatory School

3 years 9 months

Impact Project Researcher

March 2025 - June 2025 (4 months)

Kirkland, Washington, United States

- Investigated and recommended ethical guidelines for the use of realistic generative AI technology in entertainment, especially livestreaming.
- Created an AI that fully controls and mimic a livestreamer through compartmentalized gameplay, text content generation, and text-to-voice AI models in a team of three
- Presented to over 400 students and faculty in informative yet humorous way

Student Teaching Assistant

September 2023 - June 2025 (1 year 10 months)

Assisted teachers in lectures and work time with R programming, mathematical, and physics expertise, teaching 2 trimesters of Calculus, 1 of Data Science 1, and 3 of Physics.

- Volunteered and approved to teach because of extensive expertise in programming, mathematics, and physics
- Lectured about concepts in adaptable and innovative ways to tailor education to each student, guiding them to make their own discoveries
- Helped teacher prepare presentations and demonstrations. Transcribed useful formulae and concepts during them.
- Led the class through activities on days where teachers were not present
- Spoke about concepts in adaptable and innovative ways to tailor education to each student, guiding them to make their own discoveries.

Hack@EPS Club Leader

March 2022 - June 2025 (3 years 4 months)

Kirkland, Washington, United States

- Co-founded a programming club to manage and develop new features in Eastside Preparatory School (EPS) web applications, educate students in programming, and collaborate on various student-led programming projects.
 - Led management of EPSchedule, a schedule display website used by the vast majority of 600+ students and faculty, successfully increasing interest and finding successors.
 - Developed many features including lunch displays by collaborating with peers and campus departments.
 - Troubleshooted issues in a timely manner and consistently confirmed that schedules are accurate.
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- Collaborated with the graduating past maintainer to update the website, learn its intricacies, and confirm a transfer of club leadership.
 - Grew club membership from two students to more than ten, utilizing engaging activities to sustain interest.
 - Planned and launched the first Hackathon in EPS history by coordinating with Hack@EPS members, the Hack Club organization, and school faculty, resulting in a doubling of our membership.

Independent Project Researcher

December 2024 - March 2025 (4 months)

Kirkland, Washington, United States

- Programmed a gesture-controlled drone to increase ease of controls learning and allow for longer-distance use.
- Investigated methods of gesture recognition to improve accuracy
- Investigated and employed simulators to demo project in a safer environment
- Attached add-ons using soldering and other manufacturing techniques to give drone GPS and gyro access

Designer, Programmer, Varsity Head of CAD Design, Varsity Captain

October 2021 - March 2025 (3 years 6 months)

Kirkland, Washington, United States

Varsity Captain (HS Senior), Varsity Head of CAD (HS Junior)

- Led an FTC team to create a robot from scratch mostly using custom parts, achieving a world record and school record in competition level progression, achieving top 3% of FTC robotics teams worldwide.
- Navigated disputes, work distributions, and time conflicts to achieve progress and maintain a cooperative environment.

- Often conducted outreach to Junior Varsity Robotics and First Lego League teams to teach programming, design philosophy, and CAD design, resulting in the creation of the JV Robotics CAD team.

Independent Study Researcher

March 2024 - June 2024 (4 months)

Kirkland, Washington, United States

Researched Generative 3D Modeling for an independent study

- Wrote an (as of the time) unprecedented literature review paper on Generative AI in 3D applications to compile valuable research in nonprohibitive text.
- Read many papers and explained them in simple terms
- Connected with Microsoft researchers about research results

Researcher and Developer

January 2023 - June 2024 (1 year 6 months)

Kirkland, Washington, United States

Worked in three different programming and CAD related projects with diverse clients and goals

- Designed several iterations of hardware components that housed electronics in a clean, safe, and convenient way
- Led efforts to integrate strategies and creations of both Software and Hardware teams for convenient storage, manufacturing, and a clean appearance
- Found and cleaned valuable data to train a Large Language Model
- Troubleshooted many issues that involved user access, interfacing a web application and a trained LLM

EPShow Head of Authentication

July 2023 - August 2023 (2 months)

Kirkland, Washington, United States

Explored methods to allow image storage in a database and for approved images to be put on display in televisions around campus

- Developed an authentication system for school Microsoft accounts by integrating Microsoft Azure and Django.
- Created a Django-PostgreSQL database for image storage in an album-like system.
- Coordinated and motivated a team of passionate individuals to plan and contribute to this project.

Silvestria

Co-Founder

September 2023 - June 2025 (1 year 10 months)

Kirkland, Washington, United States

- Founded a non-profit LLC that designed, built, marketed, and sold custom 3D artifacts, undergoing rigorous testing and creating many versions of items to develop a secure and convenient design.
- Conducted user research to discover and implement changes in products.
- Collaborated with the American Contract Bridge League to design and market bridge accessories to players, donating all "profits" to non-profit Bridge4Youth that teaches Bridge to disadvantaged youth.

EPSchedule

Head Developer

August 2021 - June 2025 (3 years 11 months)

Led maintenance and development of EPSchedule, a schedule display website used by the majority of the more than 600-people campus population.

Troubleshooted EPSchedule issues in a timely manner and consistently confirmed that schedules are accurate.

Co-founded a programming club to manage and develop new features in Eastside Preparatory School (EPS) web applications including EPSchedule, educate students in programming, and collaborate on student-led projects.

Planned and launched the first Hackathon in EPS history by coordinating with Hack@EPS members, the Hack Club organization, and school faculty, resulting in a doubling of our membership.

National Speech & Debate Association (NSDA)

5 years 1 month

Congress Debate Competitor

September 2024 - January 2025 (5 months)

Kirkland, Washington, United States

Public Forum Debate Competitor

January 2020 - September 2024 (4 years 9 months)

Kirkland, Washington, United States

Debated mainly in Public Forum where competitors have to debate both sides of a NSDA-chosen topic

- Competed in the 2024 National Speech and Debate Tournament in Public Forum, reaching top 113 PF debaters nationwide

- Won first place in Public Forum for the 2024 Puget Sound district tournament
- Won third place in Public Forum for the 2024 Washington state tournament
- Participated in the 2023 National Speech and Debate Tournament in supplemental events
- Regularly taught new debaters on key skills in speaking and research
- Researched many topics in depth to prepare arguments, including the use of interviews

SummerThink

Founder, Manager, and Tutor

June 2024 - September 2024 (4 months)

Kirkland, Washington, United States

Founded a free tutoring program that aimed to teach four different subjects (CAD, Precalculus, Python, and Data Science) to Middle School and High School students who may not be able to access similar resources.

- As CAD course lead, created curriculum and organized team efforts to contribute materials
- Personally taught 16 sessions in CAD, Precalculus, and Python
- Set manageable and safe deadlines for each milestone and class session
- Acted as manager of the program, scheduling regular meetings with each volunteer to monitor progress
- Led by example, publishing personal schedules and teaching materials to inspire others
- Created and managed Google Classrooms and Teams Meetings
- Troubleshooted all technical issues during sessions

Google

Intern

June 2024 - September 2024 (4 months)

Seattle, WA

- Designed and implemented a Bluetooth Low Energy alternative to Google's Wi-Fi-based XDTK system, enabling Android devices to function as low-latency controllers for Unity applications under stricter bandwidth and reliability constraints, as well as in Wi-Fi-less situations.
- Engineered a resilient packet streaming and reconstruction protocol to prevent loss and corruption, improving communication stability across varying BLE conditions.
- Authored detailed technical documentation and system diagrams outlining design decisions, failure cases, and

iterative refinements to support future research and development.

Microsoft

Hunt the Wumpus Game Developer

November 2021 - June 2024 (2 years 8 months)

Kirkland, Washington, United States

2021-22 First Place Award, 2023-24 Third Place Award / Audience's Choice Award

Competed in the Hunt the Wumpus game development competition in the 2021-22 and 2023-24 seasons

- Programmed the enemy attacks and aiming/shooting controls within a 2D manic shooter game and 3D First-person shooter game
- Created most assets and effects used in the game with the Unity Editor and Blender
- Collaborated with passionate and skilled team members, leading planning and milestone creation to keep the team informed on deadlines
- Introduced Microsoft Planner to effectively record tasks and due dates

Hack Club

Jam Developer

June 2023 - October 2023 (5 months)

Shelburne, Vermont, United States

Communicated and collaborated among writers and developers to create an easily accessible database for Hack Club lesson plans.

- Enhanced user experience by implementing website navigation and search with Node.js and its libraries.
- Integrated tutorials written by Hack Club members into the website's formatting.
- Designed and assembled a lesson plan to teach Hack Club members how to use TinkerCAD.

Education

Carnegie Mellon University

Bachelor of Science - BS, Computer Science · (August 2025 - May 2028)

Eastside Preparatory School

· (September 2021 - June 2025)

Johns Hopkins Center For Talented Youth (CTY)

Logic · (July 2022 - July 2022)

Summit Debate

Public Forum Debate · (July 2022 - July 2022)

Carnegie Mellon University

Bachelor's Degree, Computational Science